

STAT 440 Project Proposal

Group 03: Marilla Cabrigas Blatnick, Kaylie Doyle, Carly Salazar

2025-09-04

Table of contents

Project Proposal	2
Group and Communication Plan	2
Members	2
Collaboration Tools and Cadence	2
Roles	2
Research Question	2
Primary Question	2
Target Outcome	2
Importance	3
Dataset Summary	3
Student Mental Stress and Coping Mechanisms Dataset	3
Observations and Variables	3
Candidate Predictors	3
Licensing	4
Data Dictionary (Concise)	4
Suitability for Regression	5
Linearity/plausibility, expected signal, anticipated interactions	5
Potential Data Issues	6
Risks and Mitigations	6
Ethical Use and Reproducibility	6
Privacy, Bias, Fairness Considerations	6
Reproducible Workflows	6
Planned Analytic Path	6
Package Setup	6
Reproducibility Snapshot	7

Project Proposal

Group and Communication Plan

Members

Marilla Cabrigas Blatnick | muhriluh@unm.edu

Kaylie Doyle | kdoyle54@unm.edu

Carly Salazar | csalazar8@unm.edu

Collaboration Tools and Cadence

Version Control | <https://github.com/carly-salazar/stat440-project>

Initial Documentation | Google Drive

Google Drive will be used for all initial brainstorming and planning. Once the analysis begins, R code and other important files will be stored in a Git repository to ensure everyone has access to the most current versions.

Roles

All team members will be responsible for contributing to each task. The roles are:

- **Analysis:** Responsible for data collection, cleaning, and interpretation.
- **Writing:** Responsible for documenting analysis, methods, and conclusions.
- **QA:** Responsible for reviewing and verifying all work for accuracy and completeness.

Research Question

Primary Question

How well can stress levels, coping mechanisms, and demographic factors predict GPA among college students? (Predictive) Which features have the most positive effect on GPA? (Explanatory)

Target Outcome

The target outcome of this research proposal is to better understand the factors that lead to an academically successful student. The GPA of an individual student will be the response variable that will measure the academic success of a student.

Importance

This data can be used to better understand which traits are associated with students that have high academic success. This could be used to identify ways to improve their student body's academic success by implementing programs. For example, if a common factor of academic success are family issues, they could provide accessible programs to cope with family issues.

Dataset Summary

Student Mental Stress and Coping Mechanisms Dataset

<https://www.kaggle.com/datasets/salahuddinahmedshuvo/student-mental-stress-and-coping-mechanisms>

Observations and Variables

This data set has a total of 760 observations ($n=760$) and 12 variables ($p=12$).

Candidate Predictors

Quantitative

- Age
- Sleep Hours
- Work/Study Hours
- Academic Pressure
- Study Load
- Financial Stress
- Family Issues
- Stress Level

Categorical

- Gender
- Degree
- Study Year
- Coping Mechanism (exercise, hobbies, socializing, etc.)

Licensing

This data set has a [Attribution 4.0 International \(CC BY 4.0\)](#), which means we need to credit the original source when we present our work.

Data Dictionary (Concise)

Variable	Type	Description	Units	Expected role (Y / X)
GPA	Quantitative (numeric, continuous)	Student grade point average, main focus of the study	Grade point average	Y
Age	Quantitative	Student's age	Years	X
Sleep Hours	Quantitative	Average hours of sleep per night	Hours/night	X
Work/Study Hours	Quantitative	Hours per day spent working or studying	Hours/day	X
Academic Pressure	Quantitative (self-reported scale)	Perceived academic pressure	1–10 scale	X
Study Load	Quantitative (self-reported scale)	Perceived workload intensity	1–10 scale	X
Financial Stress	Quantitative (self-reported scale)	Quantitative (self-reported scale)	1–10 scale	X
Family Issues	Quantitative (self-reported scale)	Stress caused by family problems	1–10 scale	X
Stress Level	Quantitative (self-reported scale)	Overall stress rating	1–10 scale	X
Gender	Categorical	Student's gender	Categories (Male, Female, Other)	X
Degree	Categorical	Academic degree enrolled (e.g., undergrad, grad)	Categories	X
Study Year	Categorical	Current year in academic program	Categories (Year 1, Year 2, etc.)	X

Variable	Type	Description	Units	Expected role (Y / X)
Coping Mechanism	Categorical	Primary strategy for handling stress	Categories (exercise, socializing, hobbies, none, other)	X

- —

Suitability for Regression

Linearity/plausibility, expected signal, anticipated interactions

- **Stress Level, Academic Pressure, Study Load, Financial Stress, Family Issues** -> expected **negative** association with GPA (higher stress → lower GPA).
- **Sleep Hours** -> expected **positive** association with GPA (more sleep → better GPA).
- **Study Load / Work Hours** -> ambiguous: higher study hours might raise GPA up to a point, but excessive load or long work hours may reduce performance. Expect **nonlinear** effects (diminishing returns).
- **Coping Mechanism** -> protective: some coping styles (exercise, structured study) may **buffer** stress effects on GPA.
- **Gender / Degree / Study Year** -> this could shift mean GPA (group differences), useful as covariates.

Nonlinearity to expect

- **Sleep and study hours** often show non-linear relationships with performance (e.g., too little and possibly too much sleep both matter).
- **Stress** -> GPA may be nonlinear (moderate pressure can improve performance; excessive stress harms it).

Anticipated interactions

- **Stress Level/Coping Mechanism**: coping reduces the harmful effect of stress on GPA.
- **Study Load/Sleep Hours**: heavy study + little sleep -> larger negative effect.
- **Financial Stress/Family Issues**: combined social stressors may amplify negative impact.
- **Academic Pressure/Study Load**: high pressure and high load could interact.

Potential Data Issues

GPA scale ambiguity: check whether GPA is on a 0–4 scale, 0–10, percent, or mixed (students might report differently).

Self-report error: GPA and all stress items may be self-reported (rounding, misreporting, grade inflation). And sensitive items (financial/family stress) and GPA may be missing non-randomly.

Outliers / impossible values: e.g., GPA > max scale, Sleep Hours = 0 or >24,

Collinearity: academic pressure, study load, and stress may be strongly correlated.

Risks and Mitigations

Since the data is synthetically generated from Kaggle, it's already downloaded and appears to be very clean. There are no missing values or out-of-bounds data points. While there may be time constraints, we can mitigate them with weekly progress checks.

Ethical Use and Reproducibility

Privacy, Bias, Fairness Considerations

The Kaggle dataset was synthetically generated using Mockaroo, an online tool designed to create random datasets based on user-defined parameters, meaning there is no personal identifiable information that will be used in this analysis. However, it could be possible that there is some bias introduced when synthetically generating the data.

Reproducible Workflows

We will use GitHub for version control, this will allow us to track changes and make our data and analysis publicly accessible. To ensure our results are fully reproducible, we'll set the random seed to 440540 for the entire project. We'll also provide the session information to document the specific software versions and packages used.

Planned Analytic Path

- EDA → simple regressions → candidate multiple models → diagnostics → refinement

Package Setup

```
if (!requireNamespace("pacman", quietly = TRUE)) install.packages("pacman")
pacman::p_load(tidyverse, janitor, broom, knitr, kableExtra, ggplot2)
set.seed(440540)
```

Reproducibility Snapshot

```
sessionInfo()
```

```
R version 4.5.0 (2025-04-11)
```

```
Platform: aarch64-apple-darwin20
```

```
Running under: macOS Sonoma 14.3
```

```
Matrix products: default
```

```
BLAS: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRblas.0.dylib
```

```
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK
```

```
locale:
```

```
[1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
```

```
time zone: America/Denver
```

```
tzcode source: internal
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices  utils      datasets  methods    base
```

```
other attached packages:
```

```
[1] kableExtra_1.4.0 knitr_1.50      broom_1.0.8      janitor_2.2.1  
[5] lubridate_1.9.4  forcats_1.0.0   stringr_1.5.1    dplyr_1.1.4  
[9] purrr_1.0.4      readr_2.1.5     tidyr_1.3.1      tibble_3.2.1  
[13] ggplot2_3.5.2    tidyverse_2.0.0
```

```
loaded via a namespace (and not attached):
```

```
[1] gtable_0.3.6      jsonlite_2.0.0    compiler_4.5.0     tidyselect_1.2.1  
[5] xml2_1.3.8        snakecase_0.11.1  textshaping_1.0.1  systemfonts_1.2.3  
[9] scales_1.4.0      yaml_2.3.10       fastmap_1.2.0      R6_2.6.1  
[13] generics_0.1.4    backports_1.5.0   svglite_2.2.1      pillar_1.10.2  
[17] RColorBrewer_1.1-3 tzdb_0.5.0        rlang_1.1.6        stringi_1.8.7  
[21] xfun_0.52         viridisLite_0.4.2 timechange_0.3.0    cli_3.6.5  
[25] withr_3.0.2       magrittr_2.0.3    digest_0.6.37      grid_4.5.0  
[29] rstudioapi_0.17.1 hms_1.1.3         lifecycle_1.0.4    vctrs_0.6.5  
[33] evaluate_1.0.3    glue_1.8.0        farver_2.1.2       pacman_0.5.1  
[37] rmarkdown_2.29    tools_4.5.0       pkgconfig_2.0.3    htmltools_0.5.8.1
```